

OptaDOS: User Guide

Version 1.3.388

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Further information about OPTADOS may be obtained from:

www.optados.org

Nota bene: OPTADOS, optados or OptaDOS, but never OPTADOS.

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Contributors

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- Angela F. Harper, University of Cambridge: Mizoguchi chemical shift for coreloss calculations.

Chapter 1

Introduction

1.1 Background

OPTADOS is a code for calculating optical, core-level excitation spectra along with full, partial and joint electronic density of states (DOS). The code was developed by merging the **LinDOS** code of Andrew Morris and Chris Pickard at University College London with the optical properties code of Rebecca Nicholls and Jonathan Yates at Oxford University. OPTADOS is written in Fortran 95 and may be run in parallel using **MPI**. At present OPTADOS interfaces with CASTEP and ONETEP output files, although it is extendible to perform calculations on any set of band eigenvalues and their derivatives generated by any electronic structure code.

The code is freely available through the GPL licence with the request that the following citation (quoted in full) is required in any publication resulting from the use of OPTADOS.

*Andrew J. Morris, R. J. Nicholls, C. J. Pickard and Jonathan R. Yates, OPTADOS: A Tool for Obtaining Density of States, Core-level and Optical Spectra from Electronic Structure Codes, Comp. Phys. Comm. **185**, 5, 1477,(2014)*

Further information and examples can be found at

www.optados.org

1.2 Features

OPTADOS generates optical, core-level excitation spectra along with full, partial and joint electronic DOS. The DOS, PDOS and JDOS take advantage of the linear and adaptive broadening schemes which are more accurate than standard Gaussian broadening since they exploit knowledge of the gradients of the bands at each k -point in the Brillouin zone. These DOS are the basis of the more advanced functionality of OPTADOS, the core and optical spectra.

Along with data text files OPTADOS also generates **.agr** files of results to be read by **grace**.

Chapter 2

Theoretical Background

2.1 Integrals over the Brillouin Zone

Many energy-resolved spectral properties of a material take the form of an integration of some function, $F_{nm}(\mathbf{k})$ in reciprocal space over the BZ; where $F_{nm}(\mathbf{k})$ are the elements of a periodic operator which commutes with the crystal translational symmetry. Perhaps the two simplest examples of integrals, hereinafter referred to as type (a) and type (b), take the form at $T = 0$ K of,

$$I^{(a)}(E) = \sum_n \int_{\text{BZ}} \frac{d\mathbf{k}}{(2\pi)^3} F_{nn}(\mathbf{k}) \delta(E_{n\mathbf{k}} - E), \quad (2.1)$$

and,

$$I^{(b)}(E) = \sum_n^{\text{occ}} \sum_m^{\text{unocc}} \int_{\text{BZ}} \frac{d\mathbf{k}}{(2\pi)^3} F_{nm}(\mathbf{k}) \delta(E - (E_{m\mathbf{k}} - E_{n\mathbf{k}})), \quad (2.2)$$

where $E_{n\mathbf{k}}$ are the energy eigenvalues of the system. They are also extendible to integration over spin-polarised channels in a straightforward way.

2.2 Brillouin Zone Sampling Schemes

Since the crystal momentum, $\mathbf{k}$, is a continuous quantum number the band structure in an *ab initio* calculation is approximated using specific k -points within the first Brillouin zone. These k -points may be chosen through symmetry perhaps using a Monkhorst-Pack grid [11] or the multi- k -point generalisation to the Baldereschi scheme [17, 12].

Obtaining the energy bands from a set of eigenenergies at discrete k -points is non-trivial. This simplest way of assigning bands, by counting eigenvalues at k -points, implicitly assumes that there are no band crossings in the BZ. This is further complicated by “band kissing” where bands approach, but due to small interactions between the bands breaking the degeneracy they remain continuous [2, 15]. Considering band coefficients and the overlap matrix it has been shown how to assign bands to individual eigenenergies [23, 18], however these still require a broadening scheme to obtain a DOS. Below we detail some approaches to obtaining a DOS from a set of eigenenergies.

2.2.1 Gaussian broadening

The reciprocal space is discretised into *sub-cells* each containing a k -point which contributes to the integral corresponding to the energy of the k -point at their centre. Band dispersion may be crudely approximated by smearing each sub-cell's contribution by a fixed-width Gaussian function, of some width, ω ,

$$\delta(E_{n\mathbf{k}} - E) \rightarrow \frac{1}{\omega\sqrt{2\pi}} \exp\left(-\frac{1}{2}((E_{n\mathbf{k}} - E)/\omega)^2\right). \quad (2.3)$$

This approach requires a large number of k -points to converge the DOS since when choosing ω there is a trade-off between representing the sharp features, such as van Hove singularities[21] and not introducing spurious oscillations due to the limited k -point sampling. The band crossing problem is not present since the bands are modelled as dispersionless.

Fixed-width Gaussian broadening may be chosen for OPTADOS calculations with, **broadening** : **fixed**.

2.2.2 Interpolative schemes

In the linear tetrahedron interpolative scheme the BZ is divided into roughly equal size tetrahedra with the band energy calculated at each vertex [7]. The DOS contribution of each tetrahedron is then calculated analytically by linear interpolation.

Linear interpolation suffers from the band crossing problem, since the interpolation always joins the lowest energy bands together, which has the effect of avoiding all band crossing and kissing. The method also does not describe van Hove singularities well since the interpolation is linear.

Higher-order interpolation schemes, such as the quadratic tetrahedron scheme, have been proposed which converge the singularities more rapidly than the linear scheme [3, 8, 9], but these still require a high-density k -point grid to mitigate the band-crossing problem.

2.2.3 Extrapolative schemes

Extrapolating from each k -point rather than interpolating between adjacent points eliminates the band-crossing problem. Müller and Wilkins show that for free electron bands the DOS is a smooth parabola when using the extrapolative method and a rough unconverged spectrum with the interpolative approach when using the same k -point mesh [13].

The linear extrapolative scheme still does not represent van Hove singularities as well as a second-order approach. Band curvatures may be combined with Löwdin perturbation theory to extrapolate to higher order [15, 16].

Linear extrapolative broadening may be chosen for OPTADOS calculations with, **broadening** : **linear**.

2.2.4 Adaptive broadening schemes

An approximation to linear extrapolation method was proposed by Yates *et al.* [22]. Each sub-cell's contribution to the DOS is broadened by a Gaussian function whose width, ω , is proportional to the energy band gradient as its k -point. This simple approach removes the spurious oscillations of the fixed broadening whilst maintaining the sharper features. The adaptive broadening scheme was introduced in Ref [22] in the context of Wannier interpolation of BZ quantities. However, the scheme is more general than this, and with OPTADOS we apply adaptive broadening directly to the quantities obtained from the electronic structure calculation.

Adaptive broadening may be chosen for OPTADOS calculations with, `broadening : adaptive`.

2.3 Density of Electronic States

When $F_{nm}(\mathbf{k}) = 1$ the type (a) integral represents the electronic density of states (DOS) at E and the total ground-state energy, E_{gs} is obtained by integrating over the occupied energies,

$$E_{\text{gs}} = \int_0^{E_F} I^{(a)}(E; F_{nm}(\mathbf{k}) = 1) dE, \quad (2.4)$$

where E_F is the Fermi energy.

The DOS is plotted with OPTADOS using the `task : dos` command.

The partial, or projected DOS (PDOS), I_μ is obtained by projecting the DOS onto local orbitals, μ , at each atomic site[19]. In this case,

$$F_{\mu n}(\mathbf{k}) = \Re \left\{ T_{\mu n}^\dagger(\mathbf{k}) T_{\nu n}(\mathbf{k}) S_{\nu \mu}(\mathbf{k})^{-1} \right\}, \quad (2.5)$$

where $T_{\mu n}(\mathbf{k}) = \langle \Psi_n(\mathbf{k}) | \phi_\mu(\mathbf{k}) \rangle$ are the overlap matrices between LCAO (linear combination of atomic orbitals) basis, ϕ_μ and plane-wave states, Ψ_n , and $S_{\mu n}(\mathbf{k}) = \langle \phi_n(\mathbf{k}) | \phi_\mu(\mathbf{k}) \rangle$, the overlap matrices of the LCAO basis.

The PDOS is plotted with OPTADOS using the `task : pdos` command. The projection is defined using the `pdos` keyword as described in the user manual.

In the case of the type (b) integral when $F_{nm}(\mathbf{k}) = 1$, we obtain the joint-DOS, (JDOS), for frequency, ω . The JDOS is plotted with OPTADOS using the `task : jdoss` command.

2.4 Dielectric response

Core-level spectra and optical properties are related to the complex dielectric function $\varepsilon_1 + i\varepsilon_2$. Within the random phase approximation, the imaginary part of the dielectric function is

$$\varepsilon_2(\omega) = \frac{\pi e^2}{\varepsilon_0 \Omega} \frac{1}{q^2} \left| \langle \psi_{\mathbf{k}}^c | e^{-i\mathbf{q} \cdot \mathbf{r}} | \psi_{\mathbf{k}+\mathbf{q}}^v \rangle \right|^2, \quad (2.6)$$

where e is the charge on an electron, ϵ_0 is the permittivity of free space, Ω is the volume of the unit cell and $\mathbf{q}$ is the momentum transfer [4]. Using the dipole approximation, expressing $\mathbf{q}$ as $q \hat{\mathbf{u}}$ and taking the limit that $q \rightarrow 0$, ϵ_2 can be written as an integral of type (b),

$$\epsilon_2(\omega) = \frac{\pi e^2}{\epsilon_0 \Omega} I^{(b)}(\hbar\omega), \quad (2.7)$$

where,

$$F_{nm}(\mathbf{k}) = |\langle \psi_{\mathbf{k}}^c | \hat{\mathbf{u}} \cdot \mathbf{r} | \psi_{\mathbf{k}}^v \rangle|^2. \quad (2.8)$$

Most electronic structure codes do not calculate $\langle \psi_{\mathbf{k}}^c | \mathbf{r} | \psi_{\mathbf{k}}^v \rangle$ directly due to the ambiguity of a position operator in periodic cells. Rather they calculate $\langle \psi_{\mathbf{k}}^c | \mathbf{p} | \psi_{\mathbf{k}}^v \rangle$, where $\mathbf{p}$ is the momentum operator, which is trivial in reciprocal space.

Since,

$$\mathbf{p} = m\dot{\mathbf{r}} = \frac{im}{\hbar} [\hat{H}, \mathbf{r}] \quad (2.9)$$

where $\hat{H}$ is the Hamiltonian operator and m the electron mass, it follows that:

$$\langle \psi_{\mathbf{k}}^c | \mathbf{r} | \psi_{\mathbf{k}}^v \rangle = \frac{\hbar}{im(e^c - e^v)} \langle \psi_{\mathbf{k}}^c | \mathbf{p} | \psi_{\mathbf{k}}^v \rangle \quad (2.10)$$

where e^c and e^v are the energy eigenvalues of the states $|\psi_{\mathbf{k}}^c\rangle$ and $|\psi_{\mathbf{k}}^v\rangle$. Note that had we added scissor operator term to our Hamiltonian $\hat{H} = \hat{H}_{KS} + \Delta_{\text{scissor}}$ then this also appears in the conversion from momentum matrix elements to optical matrix elements.

Once ϵ_2 is known, it is possible to use the Kramers-Kronig relations to obtain ϵ_1 and hence obtain an expression for the full dielectric function. The dielectric function is a tensor and in OPTADOS it is either possible to output the full dielectric tensor or calculate dielectric function for the isotropic average of $\mathbf{q}$, a particular direction of $\mathbf{q}$ or the average of $\mathbf{q}$ over a plane perpendicular to a given direction. Once the dielectric function has been calculated, core-level spectra and optical properties can be calculated.

Core-level and low-loss EELS spectra are all related to the loss function, which is defined as [5]:

$$\Lambda = \Im \left(\frac{-1}{\epsilon_1 + i\epsilon_2} \right). \quad (2.11)$$

2.4.1 Core-level spectra

For core-level spectroscopy (core-loss EELS and x-ray absorption near edge spectroscopy (XANES)) the incident perturbation is far from resonance and since $\epsilon_1 \approx 1$ and $\epsilon_2 \gg 1$ equation 2.11 reduces to $\Lambda = \epsilon_2$ [5]. The type (b) integral in equation 2.7 becomes a type (a) integral as the energy of the core state is fixed. Emission spectra (transitions from valence to core XES) can also be calculated by considering a sum over the occupied rather than unoccupied states. Core-level spectra are calculated in OPTADOS using the `task : core` command.

Several different experimental geometries are possible for core-level experiments and OPTADOS allows the calculation of spectra with $\mathbf{q}$ in a particular direction or an isotropic average.

In core-level absorption spectra there are several sources of broadening coming from the experimental set up and lifetime effects. These can be included in OPTADOS by broadening the theoretical spectrum using a combination of Gaussian and Lorentzian functions. To include instrumentation and lifetime effects, `core_LAI_broadening` should be set to true.

2.4.2 Optical properties

In the low-loss EELS regime, the approximations used for core-level spectroscopy do not hold and the full form of the loss function needs to be calculated. This is done by calculating ε_2 using equations 2.7 and 2.8 and then using the Kramers-Kronig relations to find ε_1 . Once the dielectric function has been calculated, the loss-function is simulated (without local field effects) using equation 2.11. As the dielectric function has been calculated, several other optical properties (which are listed below) can also be computed [4].

Conductivity, in units of Sm^{-1} , is computed using:

$$\sigma_1 = \varepsilon_0 \varepsilon_2 \omega, \quad (2.12)$$

$$\sigma_2 = \varepsilon_0 (1 - \varepsilon_1) \omega. \quad (2.13)$$

The refractive index is calculated using:

$$N = n + ik \quad (2.14)$$

where n and k are obtained from,

$$n^2 = \frac{1}{2}([\varepsilon_1^2 + \varepsilon_2^2]^{\frac{1}{2}} + \varepsilon_1) \quad \text{and} \quad k^2 = \frac{1}{2}([\varepsilon_1^2 + \varepsilon_2^2]^{\frac{1}{2}} - \varepsilon_1). \quad (2.15)$$

The absorption coefficient, (in units of m^{-1}), is computed using

$$\eta = \frac{2k\omega}{c}, \quad (2.16)$$

and reflection coefficient, R , from,

$$R = \frac{(n - 1)^2 + k^2}{(n + 1)^2 + k^2}. \quad (2.17)$$

All optical properties are calculated in OPTADOS using the `task : optics` command.

2.4.3 Intraband term

For metals it is necessary to consider an intraband contribution to the optical response, i.e. a term when $n = m$ in equation 2.2 [4, 1]. The contribution to ε can be written as a type (a) integral with $\hbar\omega$ set to E_F ,

$$\varepsilon_1^{\text{intra}}(\omega) = 1 - \frac{e^2}{\varepsilon_0 \Omega} \frac{1}{(\omega^2 + \Gamma^2)} I^{(a)}(E_F), \quad (2.18)$$

$$\varepsilon_2^{\text{intra}}(\omega) = \frac{e^2}{\varepsilon_0 \Omega} \frac{\Gamma}{\omega(\omega^2 + \Gamma^2)} I^{(a)}(E_F), \quad (2.19)$$

where Γ denotes the relaxation rate.

Chapter 3

Getting Started

3.1 Installation

OPTADOS is usually obtained in a gzipped tarball, `optados-X.X.tar.gz`. Extract this (`tar -xzf optados-X.X.tar.gz`) in the desired directory. Inside the `optados/` directory are a number of sub directories, `documents/`, `examples/` and `tools/`. The code may be compiled using the `Makefile` in the `optados/` directory. The `SYSTEM`, `BUILD`, `COMMS_ARCH` and `PREFIX` flags must be set, either in the `makefile.system`, or from the command line (for example `make BUILD=fast`).

3.1.1 SYSTEM

Choose which compiler to use to make OPTADOS. The valid values are:

- `gfortran` (default)
- `ifort`
- `nag`
- `pgf90`
- `A00C`
- `oneAPI`

The following are no longer supported but are provided for legacy systems:

- `g95`
- `pathscale`
- `sun`

3.1.2 BUILD

Choose the level of optimisations required when making OPTADOS. The valid values are:

- **fast** (default) All optimisations
- **debug** No optimisations, full debug information

3.1.3 COMMS_ARCH

Whether to compile for serial or parallel execution. The valid values are:

- **serial** (default)
- **mpi**

3.1.4 PREFIX

Choose where to place the OPTADOS binary. The default is the OPTADOS directory.

3.2 Usage

`optados.x86_64 [seedname]`

- **seedname**: If a seedname string is given the code will read its input from a file `seedname.odi`. The default value is `CASTEP`.

Chapter 4

Structure of the Program

The schematic structure of the program is outlined in Fig. 4.1. Each rounded-cornered box represents a Fortran95 module and the lines represent module dependencies. Modules only use data and subroutines from lower modules in the diagram. The modules are grouped into functional, structural, utility and low level. Low level modules set up the environment for communication in serial or parallel between nodes and to disk. The utility modules are not context dependent on the problem, and contain general algorithms and an interface to write **xmgrace** input files. Structural modules define data structures, hold data and perform low-level operations on them. The functional modules perform the higher-level data operations and generate output.

A description of the purpose of each module is given below.

- **algorithms**: subroutines not specific to electronic structure
- **cell**: real and reciprocal space data manipulation and storage, (*e.g.* cell vectors and k -points)
- **comms**: interfaces to parallel libraries
- **constants**: definition of constants and conversion factors
- **core**: perform core-level calculation
- **dos**: perform DOS calculation
- **dos_utils**: perform type (a) integral evaluation
- **electronic**: read electronic eigenvalue data. Store electronic data variables
- **io**: error handling, timing, and input and output units
- **jdos**: perform JDOS calculation
- **jdos_utils**: perform type (b) integral evaluation
- **optados**: main program

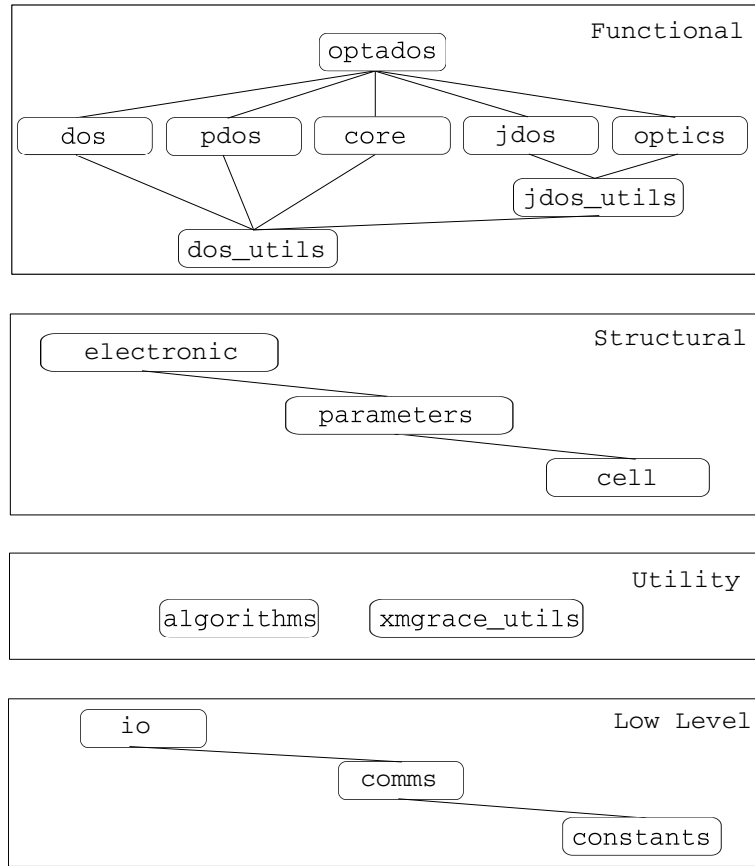


Figure 4.1: Schematic structure of the program

- `optics`: calculate dielectric function then generate optical properties
- `parameters`: read, store and check input file parameters
- `pdos`: perform PDOS calculation
- `pdis`: perform a projected dispersion calculation
- `projection_utils`: routines for reading and writing projectors
- `xmgrace_utils`: routines to write out data in xmgr format

Chapter 5

Parameters

5.1 seedname.odi File

The OPTADOS input file `seedname.odi` has a flexible free-form structure.

The ordering of the keywords is not significant. Case is ignored (so `smearing_width` is the same as `Smearing_Width`). Characters after `!`, or `#` are treated as comments. Most keywords have a default value that is used unless the keyword is given in `seedname.odi`. Keywords may be set in any of the following ways

```
smearing_width = 0.4
smearing_width : 0.4
smearing_width 0.4
```

A logical keyword can be set to `.true.` using any of the following strings: `T`, `true`, `.true..`

5.2 General Parameters

5.2.1 `character(len=20) :: task`

Tells the code what to compute.

The valid options for this parameter are:

- `dos` (default)
- `compare_dos`
- `compare_jdos`
- `jdos`
- `pdos`
- `pdispersion`
- `optics`
- `core`
- `all`

Several tasks can be specified *e.g.* to compute dos and jdost use `task : dos jdost`. However, the `compare_dos` and `compare_jdos` tasks can only be combined with each other, and no additional tasks. `compare_dos` and `compare_jdos` calculate the DOS and JDOS respectively using all broadening schemes. It is good practice to check the quality of the underlying DOS before other tasks are requested.

5.2.2 `character(len=50) :: broadening`

Specifies the scheme used to broaden a discrete sampling of the Brillouin Zone to a continuous spectral function.

The valid options for this parameter are:

- `adaptive` (default)
- `fixed`
- `linear`
- `quad` (not currently implemented)

5.2.3 integer :: iprint

This indicates the level of verbosity of the output from 1, the bare minimum: 2, with progress reports: to 3, which corresponds to full debugging output.

The default value is 1.

5.2.4 character(len=20) :: energy_unit

The energy unit to be used for writing quantities in the output files.

The valid options for this parameter are:

- eV (default)
- Ry
- Ha

5.2.5 logical :: legacy_file_format

- TRUE Read CASTEP input compatible with versions < 6.0.
- FALSE (Default) Read CASTEP input compatible for use with CASTEP versions 6.0+ and generated with the castep spectral task.

5.2.6 real(kind=dp) :: adaptive_smearing

Set the relative smearing in the adaptive scheme.

Default value is 0.4

5.2.7 real(kind=dp) :: fixed_smearing

Smearing width for fixed broadening.

If `spectral_scheme = fixed` default value is 0.3eV.

5.2.8 real(kind=dp) :: linear_smearing

Smear the linear broadening with a Gaussian of this width.

Default value is 0.0.

5.2.9 `character(len=20) :: efermi`

Choose which Fermi energy to use.

The valid options for this parameter are:

- `optados` (default) OPTADOS recalculates the Fermi energy by performing a DOS calculation.
- `file` Take the value from the output of the ab-initio calculation.
- `insulator` Assume that the material is an insulator and counts filled bands to find the Fermi energy.
- `<real number>` User supplied value.

The default value is `optados`.

5.2.10 `character(len=20) :: output_format`

Format in which to output data.

The valid options for this parameter are:

- `gnuplot`
- `grace` (default)

5.2.11 `logical :: finite_bin_correction`

Force each Gaussian to be larger than a single energy bin. (Useful for adaptive smearing and semi-core states when `numerical_intdos=TRUE`).

Default value `TRUE`.

5.2.12 `logical :: numerical_intdos`

Calculate the integrated dos by numerical integration instead of semi-analytically. (Useful for comparison with `LinDOS`.)

Default value `FALSE`.

5.2.13 `logical :: hybrid_linear`

Switch from linear broadening scheme to adaptive broadening when band gradient less than `hybrid_linear_grad_tol`. This allows for a good description of very flat bands such as

defect and semi-core states. May also be used in conjunction with `finite_bin_correction` further improving the DOS and band energy

Default value `FALSE`.

5.2.14 `real(kind=dp) :: hybrid_linear_grad_tol`

Tolerance for switching from linear to adaptive broadening when using `hybrid_linear` option.

The default value is $0.01\text{eV}/\text{\AA}$.

5.2.15 `character(len=50) :: devel_flag`

Not a regular keyword. Its purpose is to allow a developer to pass a string into the code to be used inside a new routine as it is developed.

No default.

5.3 DOS Parameters

5.3.1 `logical :: compute_band_energy`

Compute the band energy by summing bands both using CASTEP's eigenvalue and OPTA-DOS's density of states.

Default value `TRUE`.

5.3.2 `logical :: compute_band_gap`

Compute the optical, thermal and average band gap.

Default value `FALSE`.

5.3.3 `logical :: dos_per_volume`

Present DOS per simulation cell volume.

Default value `FALSE`.

5.3.4 `real(kind=dp) :: dos_min_energy`

Lower energy range for DOS and related properties.

Default value is 5eV below the lowest eigenvalue in the bands file.

5.3.5 `real(kind=dp) :: dos_max_energy`

Upper energy range for DOS and related properties.

Default value is 5eV above the highest eigenvalue in the bands file.

5.3.6 `real(kind=dp) :: dos_nbins`

Instead of setting a Default value `dos_spacing` the total number of DOS bins may be given. (Useful for comparison with LinDOS.)

5.3.7 `real(kind=dp) :: dos_spacing`

Resolution at which to compute the DOS and related properties.

Default value is 0.1eV

5.3.8 `logical :: set_efermi_zero`

Shift energy scales so that the Fermi energy is at 0.

Default value FALSE.

5.4 JDOS Parameters

5.4.1 `real(kind=dp) :: jdos_max_energy`

Upper energy range for JDOS and related properties.

Default value is the difference between the valence band maximum (or Fermi level) and the highest eigenvalue in the bands file.

5.4.2 `real(kind=dp) :: jdos_spacing`

Resolution at which to compute the DOS and related properties.

Default value is 0.01eV.

5.4.3 `real(kind=dp) :: scissor_op`

Value of the scissor operator.

Default value is 0eV (*i.e.* not used)

5.4.4 `real(kind=dp) :: exclude_bands`

This allows a list of bands which are NOT to be included in the JDOS or OPTICS calculation to be specified. The bands 1, 2, 4, 5, 6, for example, can be specified using `exclude_bands = 1,2,4-6`.

5.5 PDOS Parameters

5.5.1 `character :: pdos`

Defines which components to include in the pdos analysis:

- `angular` (decompose as s,p,d *etc.*)
- `species` (decompose onto atomic species C, H *etc.*)
- `sites` (decompose onto atomic sites C1, H1, H2 *etc.*)
- `species_ang` (decompose onto angular momentum channels and species Cs, Cp *etc.*)
- `C:H` (decompose onto Carbon and Hydrogen sites)
- `C1:C3:C4-C8` (decompose onto atoms C1, C2 and C4,C5,C6,C7,C8)
- `Si1(s;d)` (decompose onto 's' and 'd' channels for atom Si1)
- `sum:C1:C3:C4-C8` (decompose onto atoms C1, C2 and C4,C5,C6,C7,C8 and combine into the single projection)
- `‘‘C:Exi’’:H` (decompose only onto site labelled C:Exi in `seedname-out.cell` and Hydrogen sites)
- `‘‘C’’:H` (decompose only onto Carbon sites that *are not labelled* in `seedname-out.cell` and Hydrogen sites. *N.B* this is the same as `C:H` if no sites are labelled.)

5.6 Projected Dispersion Parameters

5.6.1 `character :: pdispersion`

Follows `pdos` in specifying which projectors are included in the projected dispersion calculation. Note that this task is not compatible with any options other than `EFERMI: FILE` (which is the default when `pdispersion` has been requested).

5.7 Optics Parameters

5.7.1 `character(len=20) :: optics_geom`

Specifies the geometry for the optics calculation. Possible options:

- `polycrystalline` (Isotropic average)
- `polarized`
- `unpolarized`
- `tensor` (Full dielectric tensor)

The default is `polycrystalline`.

5.7.2 `real(kind=dp) :: optics_qdir(3)`

Direction of polarisation. Must be specified if `optics_geom = polarized` or `optics_geom = unpolarized`. There is no default value

5.7.3 `logical :: optics_intraband`

If true, the intraband contribution to the dielectric function will be calculated. (Important for metals.)

The default is `FALSE`.

5.7.4 `real(kind=dp) :: optics_drude_broadening`

Value of broadening included in the Drude term expressed in s^{-1} .

The default value is `1E-14`.

5.7.5 `real(kind=dp) :: optics_lossfn_broadening`

FWHM of Gaussian used to broaden the loss function.

The default value is `0` (*i.e.* no broadening is used).

5.8 Core-level Parameters

5.8.1 `character(len=20) :: core_type`

Determines if we want absorption (transition from core to conduction ELNES / XANES) or emission (transition from valence to core XES). It is also possible to plot both.

- `absorption` (default)
- `emission`
- `all`

5.8.2 `character(len=20) :: core_geom`

Specifies the geometry for the core-spectra calculation. Possible options:

- `polycrystalline` (Isotropic average)
- `polarized`

The default is polycrystalline.

5.8.3 `real(kind=dp) :: core_qdir(3)`

Direction of polarisation. Must be specified if `core_geom = polarized`.

This should be specified as three real numbers, *e.g.* 1 0 0.

There is no default value.

5.8.4 `logical :: core_LAI_broadening`

Include life-time and instrumentation broadening.

The default is FALSE.

5.8.5 `real(kind=dp) :: LAI_gaussian_width`

FWHM of Gaussian function used to broaden spectrum.

The default value, if `core_LAI_broadening = true`, is 0 (*i.e.* no Gaussian used).

5.8.6 `real(kind=dp) :: LAI_lorentzian_width`

FWHM of fixed Lorentzian function used to broaden spectrum.

The default value, if `core_LAI_broadening = true`, is 0 (*i.e.* no fixed Lorentzian used).

5.8.7 `real(kind=dp) :: LAI_lorentzian_scale`

Variation of Lorentzian function with energy i.e. the width of the Lorentzian is (energy above the Fermi level x `core_lorentzian_scale`). If set to zero, no energy dependent broadening is included. If `core_lorentzian_scale` and `core_lorentzian_width` are both specified, the total width of the Lorentzian used will be `core_lorentzian_width` + (energy above the Fermi level x `core_lorentzian_scale`).

The default value, if `core_LAI_broadening = true`, is 0.1.

5.8.8 `real(kind=dp) :: LAI_lorentzian_offset`

Energy (in eV) above the Fermi level that the energy dependent broadening starts.

The default value, if `core_LAI_broadening = true`, is 0.

Chapter 6

CASTEP Examples

Each example in the `examples/` directory contains example CASTEP input files and a sample OPTADOS input file. To keep the OPTADOS distribution light CASTEP output files for the examples have not been provided. You need to run CASTEP on the CASTEP input files before running OPTADOS on the examples.

6.1 Density of States

- *Outline* : This is a simple example of using OPTADOS for calculating electronic density of states of crystalline silicon in a 2 atom cell.

- *Input Files*:

- `examples/Si2_DOS/Si2_dos.cell` - The CASTEP cell file containing information about the simulation cell.
- `examples/Si2_DOS/Si2_dos.param` - The CASTEP param file containing information about the parameters for the SCF and spectral calculations.
- `examples/Si2_DOS/Si2_dos.odi` - The OPTADOS input file, containing the parameters necessary to run OPTADOS.

1. Perform a CASTEP calculation on the bulk silicon using the `Si2_dos.cell` and `Si2_dos.param` input files.

```
$ castep Si2
```

This should take a couple of seconds to run. More help can be found in the tutorials on the CASTEP website www.castep.org.

2. Perform an OPTADOS calculation. Add `LEGACY_FILE_FORMAT : true` in the `Si2_DOS.odi` input file, if the CASTEP version you are using is before 6.0. Then execute:

```
$ optados.<SYSTEM>.<BUILD>.<COMMS_ARCH>.x86_64 Si2
```

This generates 3 files:

- `Si2.odo` – OPTADOS general output file.
 - `Si2.adaptive.dat` – The adaptive broadened DOS raw output data.
 - `Si2.adaptive.agr` – The adaptive broadened DOS in a file suitable to be plotted by `xmgrace`.
3. Open the `Si2.odo` file in a text editor (*e.g.* `vi` or `emacs`). OPTADOS has performed a Density of States calculation.

```
+----- Fermi Energy Analysis -----+
| From Adaptive broadening              |
| Spin Component : 1 occupation between 3.99961 and 4.00003 <- 0c |
| Spin Component : 2 occupation between 3.99961 and 4.00003 <- 0c |
| Fermi energy (Adaptive broadening) : 5.4109 eV <- EfA |
+-----+
```

It has used the integrated DOS to work out the Fermi level, and has suggested the error in the integration by indicating the number of electrons at the Fermi level. Since we had 4 up electrons and 4 down in the input file this analysis seems satisfactory.

```
+----- Electronic Data -----+
| Number of Bands : 23 |
| Grid size : 10 x 10 x 10 |
| Number of K-points : 110 |
| Spin-Polarised Calculation : True |
| Number of up-spin electrons : 4.00 |
| Number of down-spin electrons : 4.00 |
+-----+
```

Since we had `efermi : optados`, OPTADOS sets the internal value of the Fermi level to the one it has derived from the DOS. This is important for subsequent calculations. Other valid options are `file`, where OPTADOS uses the value calculated by the electronic structure code that generated the eigenvalues; `insulator`, where OPTADOS uses a value calculated from assuming the system is non-metallic; or a value set by the user.

OPTADOS now performs some analysis of the DOS at the Fermi level,

```
+----- DOS at Fermi Energy Analysis -----+
| Fermi energy used : 5.4109 eV |
| From Adaptive broadening |
| Spin Component : 1 DOS at Fermi Energy : 0.0011 eln/cell <- DEA |
| Spin Component : 2 DOS at Fermi Energy : 0.0011 eln/cell <- DEA |
+-----+
```

From this we may assume that there is a band gap.

Importantly, then OPTADOS calculates the band energy from the DOS is has calculated.


```

+----- Band Energy Analysis -----+
|          Band energy (Adaptive broadening) :          1.3609 eV          <- BEA |
|          Band energy (From CASTEP) :          1.3622 eV          <- BEC |
+-----+

```

As the quality of the OPTADOS calculation is increased these two values should converge to the same answer.

Finally OPTADOS shifts the Fermi level to 0 eV, for the output files.

4. The DOS is outputted to `Si2.adaptive.dat`. This contains 5 columns as described in the header of the file:

```

#####
#
#          O p t a D O S   o u t p u t   f i l e
#
#   Density of States using adaptive broadening
#   Generated on 12 Feb 2012 at 16:50:37
# Column      Data
#   1          Energy (eV)
#   2          Up-spin DOS (electrons per eV)
#   3          Down-spin DOS (electrons per eV)
#   4          Up-spin Integrated DOS (electrons)
#   5          Down-spin Integrated DOS (electrons)
#
#####

```

This file can be plotted by your favourite graph-plotting software. However, OPTADOS has made things easy and generated a `Si2.adaptive.agr` file which is directly plottable using `xmgrace` as shown in Fig. 6.1.

```
$ xmgrace Si2.adaptive.agr.
```

5. We now try again with a better sampling of the DOS, by setting `DOS_SPACING : 0.001` and also analyse the band gap, by setting `COMPUTE_BAND_GAP : true`. You can remove all of the OPTADOS output files by using `./tools/optados_clean` in your working directory. If you have a parallel version of OPTADOS compiled, now might be the time to try it out, if not, the serial version will be fine, but just take a bit longer. You can set `IPRINT : 2` to see a progress report in `Si2.odo`. In parallel:

```
$ mpirun -np <nprocs> optados.SYSTEM.BUILD.COMMS_ARCH.x86_64 Si2
```

but your MPI implementation may be different.

6. In `Si2.odo` we now have a new section analysing the band gap in various ways.

```

+----- Bandgap Analysis -----+
|          Number of kpoints at          VBM          CBM          |
|          Spin :   1   :           1           1           |
|          Spin :   2   :           1           1           |
+-----+

```

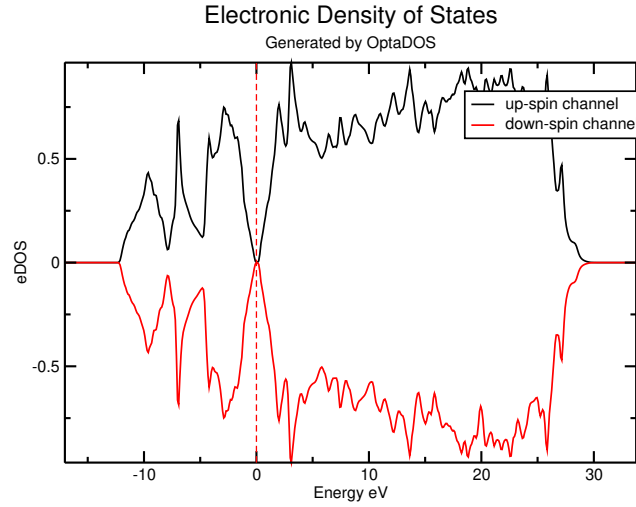


Figure 6.1: Density of States of Silicon generated by adaptive broadening and a very coarse energy sampling of 0.1 eV.

Thermal Bandgap :	0.6676272107	eV	<- TBg
Between VBM kpoint :	0.05000	0.05000	0.05000
and CBM kpoint:	-0.45000	-0.05000	-0.45000
==> Indirect Gap			

Optical Bandgap			
Spin : 1 :	2.5542517447	eV	<- OBg
Spin : 2 :	2.5542463024	eV	<- OBg
Number of kpoints with this gap			
Spin : 1 :	1		
Spin : 2 :	1		

Average Bandgap			
Spin : 1 :	3.8121372691	eV	<- ABg
Spin : 2 :	3.8121342659	eV	<- ABg
Weighted Average :	3.8121357675	eV	<- wAB

OPTADOS is very careful in its band gap analysis. It uses the bare eigenvalues (unbroadened) and works out the nature and size of the thermal gap, optical gap and the average gap over all of the Brillouin zone. In cases of multi-valleyed semiconductors OPTADOS will report the number of conduction band minima or valence band maxima with identical energies, but will not report the nature of the gap.

Increasing the number of integration points has improved the band energy of the adaptive smearing:

Band energy (Adaptive broadening) :	1.3623	eV	<- BEA
-------------------------------------	--------	----	--------

7. Now set `TASK : compare_dos` and re-run OPTADOS. OPTADOS will calculate DOS using all the broadening methods, this is good practice to see whether the broadening widths are appropriate before more advanced tasks are carried out, such as Joint-DOS, core and optical calculations.

Plotting the linear broadened DOS over the adaptive we see that the default adaptive broadening is appropriate,

```
xmgrace Si2.adaptive.agr -nxy Si2.linear.dat
```

Linear broadening, although a massive improvement over fixed broadening, sometimes appears noisy if used with very low numbers of k-points. Linear and adaptive DOS should be compared and `ADAPTIVE_SMEARING` may be tuned by eye until the adaptive DOS contains the features of the linear DOS, but with less noise. Adding a random shift to a k-point mesh can greatly increase the quality of the DOS, especially if the k-point set contained the Γ -point. It generally pays off in computational time to have a coarse mesh at low symmetry points, than a fine mesh centred on high symmetry points.

Both fixed and adaptive broadening can fail to plot the sharpest features, such as semi-core states, if the bin widths are too broad. These sharp features may be forced to be present using the narrowest Gaussian still reproducible by the chosen bin widths by setting `FINITE_BIN_CORRECTION : true`.

Linear broadening may also be improved by using `HYBRID_LINEAR : true`. Van Hove singularities and other sharp features are now described at the adaptive broadening level if the bands are flatter than `HYBRID_LINEAR_GRAD_TOL`. Hybrid linear may also take advantage of the finite bin correction if required.

8. Compare the fixed and adaptive DOS and see the advantage of adaptive broadening over standard Gaussian smearing.

6.2 Projected Density of States

We assume the reader is familiar with the previous section on Density of States calculations and is now familiar with choosing broadening widths, and running OPTADOS.

- *Outline* : This is a simple example of using OPTADOS for calculating electronic density of states of 2 atoms of crystalline silicon projected onto LCAO basis states.
- *Input Files*:
 - `examples/Si2_PDOS/Si2_dos.cell` - The CASTEP cell file containing information about the simulation cell.
 - `examples/Si2_PDOS/Si2_dos.param` - The CASTEP param file containing information about the parameters for the SCF and spectral calculations.
 - `examples/Si2_PDOS/Si2_dos.odi` - The OPTADOS input file, containing the parameters necessary to run OPTADOS.

1. Choose a broadening scheme for the projected-DOS calculation and test using `TASK : compare_dos` as explained in the previous example. Checking that the `DOS_SPACING` is sufficiently fine for the band energies to match.
2. Once the DOS looks suitable, switch to `TASK : pdos`. We choose to decompose the DOS into angular momentum channels (`PDOS : angular`) and as in the previous example we choose to recalculate the Fermi level using the calculated DOS, rather than use the Fermi level suggested by CASTEP.
3. Execute OPTADOS.
4. The output can be found in `Si2.pdos.dat`.

```
#####
#
#                               O p t a D O S   o u t p u t   f i l e
#
#   Generated on 13 Feb 2012 at 10:15:10
#####
#-----+
#|                               Partial Density of States -- Projectors                               |
#-----+
#| Projector:    1 contains:                                           |
#|               Atom                AngM Channel                     |
#|               Si    1                s                             |
#|               Si    2                s                             |
#-----+
#| Projector:    2 contains:                                           |
#|               Atom                AngM Channel                     |
#|               Si    1                p                             |
#|               Si    2                p                             |
#-----+
#| Projector:    3 contains:                                           |
#|               Atom                AngM Channel                     |
#|               Si    1                d                             |
#|               Si    2                d                             |
#-----+
#| Projector:    4 contains:                                           |
#|               Atom                AngM Channel                     |
#|               Si    1                f                             |
#|               Si    2                f                             |
#-----+
```

The header shows that there are four projectors described below. The first containing the s-channels of both silicon atoms, the second the p-channels *etc.*

5. The output is easily plotted using `xmgrace`:
`xmgrace -nxy Si2.pdos.dat`

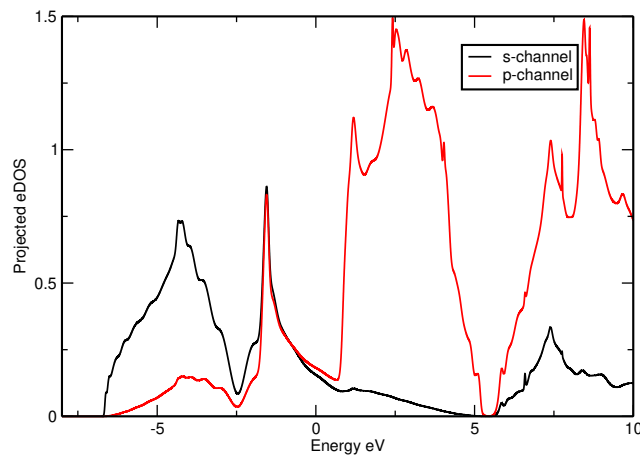


Figure 6.2: Density of States of Silicon generated by adaptive broadening projected onto LCAO momentum states.

6. Setting `DOS_SPACING` : 0.001 gives a high quality plot, as shown in Fig 6.2

7. Other things to try are:

- `PDOS : Si1;Si2(s)` – Output the PDOS on Si atom 1 and the PDOS on the s-channel of Si atom 2. (Resulting in two projectors)
- `PDOS : sum:Si1-2(s)` – Output the sum of the s-channels on the two Si atoms. (Resulting in one projector)
- `PDOS : Si1(p)` – Output the p-channel on Si atom 1. (Resulting in one projector)

6.3 Projected Bandstructure

- *Outline* : This is a simple example of using OPTADOS to calculate the electronic bandstructure of 2 atoms of crystalline silicon projected onto LCAO basis states. The process is almost identical to the PDOS example above, but requires performs the projections on a continuous k-point path rather than a grid.

- *Input Files*:

- `examples/Si2_PDIS/Si2_pdis.cell` - The CASTEP cell file containing information about the simulation cell.
- `examples/Si2_PDIS/Si2_pdis.param` - The CASTEP param file containing information about the parameters for the SCF and spectral calculations.
- `examples/Si2_PDOS/Si2_pdis.odi` - The OPTADOS input file, containing the parameters necessary to run OPTADOS.

1. Run CASTEP in the same way as previous examples, ensuring that `PDOS_CALCULATE_WEIGHTS: TRUE` is set in the CASTEP parameter file.

2. Run OPTADOS with the desired projections, following the same format as PDOS in `Si2_pdis.odi`.
3. The output can be found in `Si2_pdis.dat`, which contains the weight of each projector at each eigenvalue, for each k-point. Plotting this data in an attractive way is tricky; you might consider trying the Python package `matador` (www.matador.science).

6.4 JDOS

See `examples/Si2_JDOS/`. This is a simple example of using OPTADOS for calculating joint electronic density of states. We choose to recalculate the Fermi level using the calculated DOS, rather than use the Fermi level suggested by CASTEP and so `EFERMI: OPTADOS` is included in the `Si2.odi` file.

1. Execute OPTADOS using the example files. The JDOS is outputted to `Si2.jadaptive.dat`. A file suitable for plotting using `xmgrace` is written to `Si2.jadaptive.agr`.
2. Check the effect of changing the sampling by increasing and decreasing the value of `JDOS_SPACING` in the `Si2.odi` file.
3. If `TASK : compare_jdos` is used instead, OPTADOS will calculate the JDOS using all the broadening methods. This is good practice to see whether the broadening widths are appropriate before more advanced tasks are carried out.

6.5 OPTICS

Two sets of example files are provided for calculations of optical properties. For each example, the CASTEP files containing all the cell and simulation parameters are included, along with an OPTADOS input file. We assume that the reader is familiar with the previous sections on DOS and JDOS.

- `examples/Si2_OPTICS/` This is a simple example of using OPTADOS to calculate the optical properties of crystalline silicon, which is an insulator.
- `examples/Al_OPTICS/` This is a simple example of using OPTADOS to calculate the optical properties of a metal, aluminium.

6.5.1 Silicon

1. Choose a broadening scheme for the JDOS as explained in the JDOS example. Once the JDOS looks suitable, switch the `task` from `JDOS` to `OPTICS` and execute OPTADOS to calculate the optical properties. Several `*.dat` files are produced:
 - `Si2_OPTICS_absorption.dat` : This file contains the absorption coefficient (second column) as function of energy (first column).

- `Si2_OPTICS_conductivity.dat`: This file contains the conductivity outputted in SI units (Siemens per metre). The columns are the energy, real part and imaginary part of the conductivity respectively.
- `Si2_OPTICS_epsilon.dat`: This file contains the dielectric function. The columns are the energy and real and imaginary parts of the dielectric function respectively. The file header also includes the result of the sum rule $\int_0^{\omega'} \text{Im}\epsilon(\omega) d\omega = N_{eff}(\omega')$. N_{eff} is the effective number of electrons contributing to the absorption process, and is a function of energy.
- `Si2_OPTICS_loss_fn.dat`: This file contains the loss function (second column) as a function of energy (first column). The header of the file shows the results of the two sum rules associated with the loss function $\int_0^{\omega'} \text{Im}\left(\frac{-1}{\epsilon(\omega)}\right) \omega d\omega = N_{eff}(\omega')$ and $\int_0^{\omega'} \text{Im}\left(\frac{-1}{\epsilon(\omega)}\right) \frac{1}{\omega} d\omega = \frac{\pi}{2}$.
- `Si2_OPTICS_reflection.dat`: This file contains the reflection coefficient (second column) as a function of energy (first column).
- `Si2_OPTICS_refractive_index.dat`: This file contains the refractive index. The columns are the energy and real and imaginary parts of the refractive index respectively.

Corresponding `*.agr` files are also generated which can be plotted easily using `xmgrace`.

2. Change parameters `JDOS_SPACING` and `JDOS_MAX` and check the effect on the optical properties. Note: all of the other optical properties are derived from the dielectric function.
3. The OPTADOS input file has been set up to calculate the optical properties in the polycrystalline geometry (`optics_geom = polycrystalline`). It is possible to calculate either polarised or unpolarised geometries, or to calculate the full dielectric tensor. To calculate the full dielectric tensor set `optics_geom = tensor`. This time only the file `Si2_OPTICS_epsilon.dat` is generated. The format of this file is the same as before (the columns are the energy and the real and imaginary parts of the dielectric function respectively), but this time the six different components of the tensor are listed sequentially in the order ϵ_{xx} , ϵ_{yy} , ϵ_{zz} , ϵ_{xy} , ϵ_{xz} and ϵ_{yz} .
4. Additional broadening can be included in the calculation of the loss function. This is done by including the keyword `optics_lossfn_broadening` in the OPTADOS input file. If you include this keyword and re-run OPTADOS, you will find that the file `Si2_OPTICS_loss_fn.dat` now has three columns. These are the energy, unbroadened spectrum and broadened spectrum respectively.

6.5.2 Aluminium

1. As for the silicon example, a broadening scheme for the JDOS should first be determined.
2. Aluminium is a metal so we need to include both the interband and intraband contributions to the dielectric function. To include the intraband contribution `optics_intraband = true` must be included in the OPTADOS input file. When you run OPTADOS the same files are generated as when only the interband term is included.

The `Al_OPTICS_epsilon.dat` file has the same format as before, but it now contains sequentially the interband contribution, the intraband contribution and the total dielectric function. (The file `Al_OPTICS_epsilon.agr` only contains the interband term.) In the same way, `Al_OPTICS_loss_fn.dat` contains the interband contribution, intraband contribution and total loss function. All other optical properties are calculated from the total dielectric function and the format of the output files remains the same.

3. In the case where the dielectric tensor is calculated and the intraband term is included, only the `Al_OPTICS_epsilon.dat` file is generated. As before it contains each component, but this time it lists sequentially the interband contribution, intraband contribution and total dielectric function for each component.
4. This time, if additional broadening for the loss function is included by using the keyword `optics_lossfn_broadening`, `AL_OPTICS_loss_fn.dat` will contain four sequential data sets. These are the interband contribution, the intraband contribution, the total loss function without the additional broadening and the broadened total loss function.

6.6 CORE

See `examples/Si2_CORE/`. This is a simple example of using OPTADOS for calculating core level absorption spectra for crystalline silicon. We assume that the reader is familiar with the previous section on calculating DOS.

1. We begin by running a CASTEP calculation using the files provided in `examples/Si2_CORE`. Note that we do not specify pseudopotentials in the `Si2_CORE.cell` file hence require CASTEP to generate on-the-fly pseudopotentials. This is important as most pseudopotential formats do not contain enough information for the PAW reconstruction needed for a CORE calculation. For any atom a CORE spectra is to be calculated an On-the-fly pseudopotential must be used.

Execute OPTADOS using the OPTADOS input file provided and the file `Si2_CORE_core_edge.dat` will be created. The file contains two columns, the first is the energy and the second is the spectrum. This file contains the following edges:

```
# Si 1      K1
# Si 1      L1
# Si 1     L2,3
# Si 2      K1
# Si 2      L1
# Si 2     L2,3
```

i.e. all edges from all atoms are produced.

2. To include a core-hole in the calculation, first one atom is chosen to have the excitation. To begin, we will keep two atoms in the unit cell and distinguish one atom by changing the `%BLOCK POSITIONS_FRAC`


```

%BLOCK POSITIONS_FRAC
      Si:exi      0.0000000000      0.0000000000      0.0000000000
      Si          0.2500000000      0.2500000000      0.2500000000
%ENDBLOCK POSITIONS_FRAC

```

in the `Si2_CORE.cell` file. The atom `Si:exi` is the one to have a core-hole. To create a core-hole we remove a 1s electrons from the electronic configuration used in the generation of the pseudopotential. We have already generated an on-the-fly pseudopotential without a core-hole in the previous section. Information about the pseudopotentials is included at the top of the `Si2_CORE.castep` file.

```

=====
| Pseudopotential Report - Date of generation 16-05-2012 |
=====
| Element: Si Ionic charge: 4.00 Level of theory: LDA |
|
|               Reference Electronic Structure |
|      Orbital      Occupation      Energy |
|      3s           2.000          -0.400 |
|      3p           2.000          -0.153 |
|
|               Pseudopotential Definition |
|      Beta      l      e      Rc      scheme      norm |
|      1         0    -0.400    1.797      qc         0 |
|      2         0     0.250    1.797      qc         0 |
|      3         1    -0.153    1.797      qc         0 |
|      4         1     0.250    1.797      qc         0 |
|      loc       2     0.000    1.797      pn         0 |
|
| Augmentation charge Rinner = 1.298 |
| Partial core correction Rc = 1.298 |
=====
| "2|1.8|1.8|1.3|2|3|4|30:31:32LGG(qc=4)" |
=====
|               Author: Chris J. Pickard, Cambridge University |
=====

```

The line

```
2|1.8|1.8|1.3|2|3|4|30:31:32LGG(qc=4)
```

specifies the parameters used to create the pseudopotential. We use this as the starting point and then remove one of the core 1s electrons to create a core-hole pseudopotential. This is done by including `{1s1.00}` in the pseudopotential string as shown:

```
2|1.8|1.8|1.3|2|3|4|30:31:32LGG{1s1.00}(qc=4)
```

If, instead of removing a 1s electron, we wanted to remove a 2s electron from the core, we would have included {2s1.00} instead of {1s1.00} in the pseudopotential string.

We are only interested in the spectra from the atom with the core-hole and so copy the pseudopotential file generated by the previous calculation (Si_OFT.usp) to Si_LDA.usp. Then include

```
%BLOCK SPECIES_POT
Si:exi    2|1.8|1.8|1.3|2|3|4|30:31:32LGG{1s1.00}(qc=4)
Si        Si_LDA.usp
%ENDBLOCK SPECIES_POT
```

in the CASTEP Si2_CORE.cell file.

To maintain the neutrality of the cell, we include

```
CHARGE : +1
```

in the Si2_CORE.param file. Run the calculation. This time the Si2_CORE_core_edge.dat file will contain only the edges from the core-hole atom. Compare the K-edge from the core-hole calculation with the previous non-core-hole calculation.

3. The periodic images of the core-hole will interact with one another. As this is unphysical, we need to increase the distance between the core-holes. This is done by creating a supercell. To start with we use a face-centred unit cell rather than the primitive unit cell. This is done by changing the lattice parameters and fractional co-ordinates to:

```
%BLOCK LATTICE_CART
5.46  0.00  0.00
0.00  5.46  0.00
0.00  0.00  5.46
%ENDBLOCK LATTICE_CART

%BLOCK POSITIONS_FRAC
Si:exi    0.0000000000    0.0000000000    0.0000000000
Si        0.5000000000    0.5000000000    0.0000000000
Si        0.5000000000    0.0000000000    0.5000000000
Si        0.0000000000    0.5000000000    0.5000000000
Si        0.2500000000    0.2500000000    0.2500000000
Si        0.7500000000    0.2500000000    0.7500000000
Si        0.2500000000    0.7500000000    0.7500000000
Si        0.7500000000    0.7500000000    0.2500000000
%ENDBLOCK POSITIONS_FRAC
```

Run OPTADOS and compare the spectrum from the face-centred unit cell with that from the primitive unit cell. Continue constructing larger unit cells until the core-hole spectrum stops changing with increasing separation between the periodic images.

4. Other things to try include
 - Changing the geometry from polycrystalline to polarised
 - Including life-time and instrumentation broadening

6.7 CORE with Mizoguchi Chemical Shift

In the previous section we discussed including a core-hole in the `seedname.cell` file. In order to account for the energetic contribution of the core-hole to the absorption or emission edge users may find it useful to include the chemical shift of the energy term as described by Mizoguchi *et al.* [10]. The Mizoguchi shift, rescales the absorption or emission energy by calculating the energy difference between the ground state energy with and without the core-hole included in the input `seedname.cell` file. See `examples/Mizoguchi` for input files described here.

1. We begin by running a CASTEP calculation using the files provided in `examples/Mizoguchi`. We have included a 1s core hole on the Ge atom in the cell in order to obtain a K-edge spectrum. This is specified by adding a `Ge:exc` label in the `% BLOCK POSITIONS_FRAC` and by including a core hole in the pseudopotential string as shown here:

```
%BLOCK positions_frac
Na 0.620573953121250      0.758852093757499      0.836258492531341
Na 0.379426046878750      0.241147906242501      0.163741507468659
Na 0.220453843197638      0.559092313604724      0.600726924771292
Na 0.779546156802362      0.440907686395276      0.399273075228708
Ge:exc 0.000000000000000  0.000000000000000  0.000000000000000
%ENDBLOCK positions_frac

%BLOCK species_pot
Ge:exc 3|1.5|13|15|17|30U:31U:32:40:41(qc=6){1s1}
Na 2|1.3|14|16|18|20U:30:21(qc=7)
%ENDBLOCK species_pot
```

Once the CASTEP calculation is completed, run OPTADOS and look at the output. You will see that the energy range of the spectrum is from 0 eV to 30 eV, as we have not yet applied the Mizoguchi chemical shift term

2. Next, we want to add in the shift which accounts for the relative energy of our cell with a core-hole to that without a core hole. It is best to rename your `NaGe.cell` and `NaGe.param` files so that they are not overwritten in the next CASTEP calculation. We will now get the ground state energy of the cell without a core hole included. To do this, we will remove the specification for our pseudopotential which included a core hole so that your cell file (renamed to something like `NaGe-singlepoint.cell`) looks like this:

```
%BLOCK positions_frac
Na 0.620573953121250      0.758852093757499      0.836258492531341
Na 0.379426046878750      0.241147906242501      0.163741507468659
Na 0.220453843197638      0.559092313604724      0.600726924771292
Na 0.779546156802362      0.440907686395276      0.399273075228708
Ge 0.000000000000000      0.000000000000000      0.000000000000000
%ENDBLOCK positions_frac
```

```
%BLOCK species_pot
Ge 3|1.5|13|15|17|30U:31U:32:40:41(qc=6)
Na 2|1.3|14|16|18|20U:30:21(qc=7)
%ENDBLOCK species_pot
```

To run a singlepoint calculation, we want to remove the `CHARGE : +1` that we added in the `.param` file and change the task to `task : singlepoint` so your `NaGe-singlepoint.param` file should look like:

```
task : singlepoint
cut_off_energy : 250 eV
xc_functional : PBE
! charge : +1
write_checkpoint : None
```

Here we have commented out the line for charge inclusion, and changed the task keyword. Run the CASTEP calculation and compare the ground state total energies of your two different files. You should see that the groundstate energy of -11848.48889250 eV is less than the core hole groundstate energy.

3. Now we want to calculate the Mizoguchi chemical shift associated with this difference in energies. To do this, we will use the code located at `tools/miz_chemical_shift`. This code is written in PYTHON and you can see the different keywords it takes using the `--help` flag. To calculate the shift for our system we will use the command:

```
../../tools/miz_chemical_shift -i NaGe -e Ge -s NaGe-singlepoint
```

Or alternatively:

```
../../tools/miz_chemical_shift -i NaGe -e Ge -t -11848.48889250
```

Where we have specified the location of the `miz_chemical_shift` script, the name of the file we performed the core-hole calculation on `-i NaGe`, the element we performed our calculation on `-e Ge` and either the file with the singlepoint calculation `-s NaGe-singlepoint` or the value of the singlepoint energy without a core hole `-t -11848.48889250`. **NOTE:** The `miz_chemical_shift` script is written assuming that you are providing the CASTEP `NB est. OK energy`. If it cannot find the `NB est. OK energy` then it will look for the `Final energy`. Therefore it is important if for example your calculation includes a dispersion correction or Hubbard U parameter which affects the final ground state energy to provide the energy before that correction is applied. This way the `-t` flag will reflect the same energy as the core-hole final energy. If you use the `-s` flag and get the final ground state energy from a file, `miz_chemical_shift` will explicitly state which energy it is using in the output `mizoguchi.out` file. Run this command and take a look at your `NaGe.odi` file. You will see a line has been added:

```
core_chemical_shift : 11142.275218
```

Now when we run OPTADOS again, the energy scale on the x-axis will be shifted by the amount specified in your input file for OPTADOS . This is especially useful when comparing your results to experimental data, as it will allow you to apply a shift to your output from optados into the experimentally relevant energy range. For reference, the known experimental absorption edge for Ge is 11.1031 eV, very close to the shift calculated using the Mizoguchi method. In addition, this is a useful tool when calculating this shift across different core-hole sites in your cell, when your cell has more than one symmetry-inequivalent atom, such that the relative absorption energies of each atom are accounted for.

Chapter 7

ONETEP Examples

7.1 (L)PDOS

ONETEP outputs an OPTADOS compatible `seedname_prefix.bands` file, and from version 4.2 onwards, a `seedname_prefix.pdos_bin`, `seedname_prefix.dome_bin` and a `seedname-out.cell` file. Where `prefix` may be one of `val`, `cond` or `joint`. A normal ONETEP run will only output the valence files (`val`), if conduction states are required, a ONETEP conduction calculation should be run, in which case the `cond` and `joint` files will also be written to disk. For a full DOS (conduction and valence), the `joint` files should be used with OPTADOS.

When running ONETEP, in order to output OPTADOS compatible files, `DO_PROPERTIES` must be set in the input file or a properties calculation must be run in continuation of a prior, converged calculation.

`PDOS_OPTADOS_OUTPUT` should be set to `TRUE`, this is the default value.

`PDOS_MAX_L=integer`, where `integer` specifies the maximum angular momentum spherical wave to project onto, i.e. `s=0`, `p=1`, `d=2`, etc.

`DOS_SMEAR` should be set to a value greater than 0.0. When `DOS_SMEAR>0.0`, then the `seedname_prefix.bands` file will be written to disk, if `PDOS_MAX_L>=0`, then a PDOS calculation will be performed and the associated files (mentioned above) will also be written to disk.

When outputting the PDOS weights in `seedname_prefix.pdos_bin`, ONETEP will also output a text file called `pdos_weights.dat` which is not associated with OPTADOS but meant for manual inspection of the PDOS weights.

ONETEP does not produce an OPTADOS input file, this should be the only additional file (besides an OPTADOS binary) required to perform a ONETEP +OPTADOS (L)PDOS calculation.

All of the DOS functionality in ONETEP is also PAW compatible, though it is not mandatory.

7.2 EELS/ELNES

ONETEP can also produce OPTADOS compatible `seedname.elnes_bin` files by asking for an EELS[20] (conduction calculation) properties post-processing calculation. This is done by specifying `COND_CALC_EELS=TRUE` in the ONETEP input file.

To perform an EELS calculation, the PAW method must be used (`PAW=TRUE` specified in the input file and PAW datafiles for each species). For those species which the user wishes to probe core to conduction transitions, core wavefunction data must also be supplied.

Finally, for EELS calculations, a conduction calculation must be performed, so `COND_ENERGY_RANGE`, which specifies the energy window above the HOMO in which conduction states will be optimised and `COND_ENERGY_GAP`, which specifies the energy gap between the highest optimised state and the lowest un-optimised state must be set in the ONETEP input file.

More can be found on EELS/ELNES here: ¹

¹http://www2.tcm.phy.cam.ac.uk/onetep/pmwiki/uploads/Main/Documentation/eels_in_onetep.pdf

Chapter 8

Frequently Asked Questions

8.1 OptaDOS crashes complaining that it can't read the seed.bands or seed.cst_ome file.

Which version of CASTEP are you using? See Sect. 5.2.5.

8.2 The examples do not work and I'm using castep version 6.0

CASTEP version 6.0 was released during the transition of the spectral module and features. The easiest way to solve this is to obtain a copy of CASTEP > 6.0, where OPTADOS will work “out of the box”. However, to make the examples work with CASTEP 6.0 the OPTADOS examples require a few tweaks outlined below.

- Replace SPECTRAL_KPOINTS_MP_GRID in the cell file with BS_KPOINTS_MP_GRID and OPTICS_KPOINTS_MP_GRID.
- Remove SPECTRAL_TASK : DOS from the param file.
- Add DEVEL_FLAG:old_filename to the odi file.
- Add WRITE_CELL_STRUCTURE : true to your CASTEP cell file to generate the symmetries of the cell if you are doing pdos, core or optics calculations.
- Read Section 8.3 if you are planning to do a partial density of states calculation.

8.3 I'm getting odd results for my PDOS calculation when using castep version 6.0

The spectral module in CASTEP 6.0 generates the pdos weights commensurate with the optics grid you have chosen. However this gets overwritten by the main routines. Hence to get the

right PDOS weights you must set `PDOS_CALCULATE_WEIGHTS : FALSE` in the param file.

8.4 OptaDOS is reporting the wrong number of kpoints

This is a rare bug that can happen if you have picked the magic k-point offset for your grid, that is a $3/(4n)$ offset where n is the number of MP grid points in that direction. It is a very difficult edge-case to catch. Until a permanent workaround is found you can tell OPTADOS the right number using the `KPOINT_MP_GRID nx ny nz` in the odi file.

8.5 I'd like OptaDOS to do X as well

Contact the developers we're always interested in discussing new functionality.

8.6 I'd like to help, what can I do?

Contact the developers, there's always more functionality that we'd like to add to the code.

8.7 I think I've found a bug: what should I do?

- Check and re-check that it is a bug.
- Check the output of the electronic structure code.
- Check that you're using the latest version of OPTADOS.
- Email the developers the input and output files with `iprint : 3` and as much information about the problem as possible.

Appendix A

Interface with other codes

Currently OPTADOS is interfaced with CASTEP and ONETEP. This appendix consists of Fortran95 code which defines the input files for OPTADOS so as to aid its interfacing with other electronic structure codes. Presented below are the type declaration statements and write statements that may be used in other electronic structure codes or converters to generate the .bands, .ome_bin, .pos_bin and .elnes_bin files used as inputs to OPTADOS. These code snippets are also available in the `tools/` directory of the OPTADOS distribution.

A.1 .bands file

The .bands file is necessary for any calculation using OPTADOS it contains the energy eigenvalues at each k-point and data about the electronic properties of the system. The bands file is a formatted file.

```
integer,parameter:: dp=selected_real_kind(15,300) ! Define double precision
integer:: num_kpoints                ! Number of kpoints
integer:: num_spins                  ! Number of spins
integer:: max_eigenv                 ! Number of bands included in matrix elements
integer:: num_electrons(num_spins)   ! Number of electrons of each spin
integer:: num_eigenvalues(num_spins) ! Number of eigenvalues of each spin
real(dp):: efermi(num_spins)         ! Fermi level calculated from electronic
                                      ! structure code for each spin
real(dp):: cell_lattice(1:3,1:3)     ! Real-space lattice vectors
real(dp):: kpoint_positions(num_kpoints,1:3) ! K-point position vectors in
                                      ! fractions of Brillouin Zone
real(dp):: kpoint_weight(num_kpoints) ! K-point weight
real(dp):: band_energy(max_eigenv, num_spins, num_kpoints) ! Energy eigenvalue
                                                         ! list

write (band_unit,'(a,i4)') 'Number of k-points ', num_kpoints
write (band_unit,'(a,i1)') 'Number of spin components ', num_spins
if(num_spins>1) then
```

```

    write (band_unit,'(a,2g10.4)'),'Number of electrons ', num_electrons(:)
else
    write (band_unit,'(a,g10.4)'),'Number of electrons ', num_electrons(:)
endif
if(num_spins>1) then
    write (band_unit,'(a,2i4)'),'Number of eigenvalues ', num_eigenvalues(:)
else
    write (band_unit,'(a,i4)'),'Number of eigenvalues ', num_eigenvalues(:)
endif
if(num_spins>1) then
    write (band_unit,'(a,2f12.6)')'Fermi energy (in atomic units) ', efermi(:)
else
    write (band_unit,'(a,f12.6)')'Fermi energies (in atomic units) ', efermi(:)
endif
write(band_unit,'(a)') 'Unit cell vectors'
write(band_unit,'(3f12.6)') cell_lattice(1,:)
write(band_unit,'(3f12.6)') cell_lattice(2,:)
write(band_unit,'(3f12.6)') cell_lattice(3,:)

do ik=1,num_kpoints
    write (band_unit,'(a,i4,4f12.8)') 'K-point ', ik, &
        & kpoint_positions(num_kpoints,1:3), kpoint_weight(num_kpoints)
    do is=1,num_spins
        write(band_unit,'(a,i1)')'Spin component ',is
        do ib=1,num_eigenvalues(is)
            write (band_unit,*) band_energy(ib,is,ik)
        end do
    end do
end do
end do

```

A.2 .ome_bin file

The `.ome_bin` file is required for adaptive and linear broadening, and for optics calculations. Note that a `.dome` file is just the diagonal elements of the `.ome`. The `.ome_bin` file is an unformatted file. Note that, in line with many other electronic structure codes, OPTADOS reads in the momentum matrix elements, $\langle \psi_{\mathbf{k}}^c | \mathbf{p} | \psi_{\mathbf{k}}^v \rangle$, and converts them to OMEs internally.

```

integer,parameter:: dp=selected_real_kind(15,300) ! Define double precision
real(dp):: file_version=1.0_dp      ! File version
character(len=80):: file_header    ! File header comment
integer:: num_kpoints              ! Number of k-points
integer:: num_spins                ! Number of spins
integer:: max_eigenv               ! Number of bands included in matrix elements
integer:: num_eigenvalues(1:num_spins) ! Number of eigenvalues per spin channel
complex(dp):: optical_mat(max_eigenv,max_eigenv,1:3,1:num_kpoints,num_spins)! OMEs

```

```

write(ome_unit) file_version
write(ome_unit) file_header

do ik=1,num_kpoints
  do is=1,num_spins
    write(ome_unit) (((optical_mat(ib,jb,i,ik,is),ib=1,num_eigenvalues(is)), &
      &jb=1,num_eigenvalues(is)),i=1,3)
  end do
end do

```

A.3 .pdos_bin file

The .pdos_bin file is required for PDOS calculations. The .pdos_bin file is an unformatted file.

```

integer,parameter:: dp=selected_real_kind(15,300) ! Define double precision
integer:: num_kpoints          ! Number of k-points
integer:: num_spins            ! Number of spins
integer:: num_popn_orb         ! Number of LCAO projectors
integer:: max_eigenv           ! Number of bands included in matrix elements
real(dp):: file_version=1.0_dp ! File version
integer:: species(1:num_popn_orb)! Atomic species associated with each projector
integer:: ion(1:num_popn_orb)  ! Ion associated with each projector
integer:: am_channel(1:num_popn_orb) ! Angular momentum channel
integer:: num_eigenvalues(1:num_spins) ! Number of eigenvalues per spin channel
real(dp):: kpoint_positions(1:num_kpoints,1:3) ! k_x, k_y, k_z in fractions of BZ
real(dp):: pdos_weights(1:num_popn_orb,max_eigenv,num_kpoints,num_spins)!Matrix elements
character(len=80):: file_header ! File header comment

write(pdos_file) file_header
write(pdos_file) file_version
write(pdos_file) num_kpoints
write(pdos_file) num_spins
write(pdos_file) num_popn_orb
write(pdos_file) max_eigenv
write(pdos_file) species(1:num_popn_orb)
write(pdos_file) ion(1:num_popn_orb)
write(pdos_file) am_channel(1:num_popn_orb)

do nk=1,num_kpoints
  write(pdos_file) nk, kpoint_positions(nk,:)
  do ns = 1,num_spins
    write(pdos_file) ns
    write(pdos_file) num_eigenvalues(ns)
  end do
end do

```

```
do nb = 1,num_eigenvalues(ns)
  write(pdos_file) real(pdos_weights(num_popn_orb,nb,nk,ns))
end do
end do
end do
```

A.4 .elnes_bin file

The .elnes_bin file is required for ELNES calculations. The .elnes_bin file is an unformatted file.

```
integer,parameter:: dp=selected_real_kind(15,300) ! Define double precision
integer:: tot_core_projectors ! Total number of core states included in matrix elements
integer:: max_eigenvalues    ! Number of bands included in matrix elements
integer:: num_kpoints        ! Number of k-points
integer:: num_spins          ! Number of spins
integer:: num_eigenvalues(1:num_spins) ! Number of eigenvalues
integer:: species(1:tot_core_projectors) ! Atomic species associated with each projector
integer:: ion(1:tot_core_projectors) ! Ion associated with each projector
integer:: n(1:tot_core_projectors) ! Principal quantum number associated with each projector
integer:: lm(1:tot_core_projectors) ! Angular momentum quantum number associated with each projector
complex(dp):: elnes_mat(tot_core_projectors,max_eigenvalues,3,num_kpoints,num_spins) ! matrix elements

write(elnes_file) tot_core_projectors
write(elnes_file) max_eigenvalues
write(elnes_file) num_kpoints
write(elnes_file) num_spins
write(elnes_file) species(1:tot_core_projectors)
write(elnes_file) ion(1:tot_core_projectors)
write(elnes_file) n(1:tot_core_projectors)
write(elnes_file) lm(1:tot_core_projectors)

do nk = 1,num_kpoints
  do ns = 1, num_spins
    write(elnes_file) (((elnes_mat(orb,nb,indx,nk,ns),orb=1,&
      &tot_core_projectors),nb=1,num_eigenvalues(ns)),indx=1,3)
  end do
end do
```


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